



AMTEC
AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2024 – 0571



Agricultural Machinery – Four-Wheel Tractor SONALIKA WORLDTRAC 90RX (MST)

Test Report Number	2024 – 0571
Date of Issuance	December 3, 2024
Valid Until	December 3, 2029
Agricultural Machinery	Four-Wheel Tractor
Type	Four-wheel drive, Four-cylinder, Turbo-charged diesel engine, 12F + 12R transmission system, Full hydrostatic steering system
Brand	SONALIKA
Model	WORLDTRAC 90RX (MST)
Test Requested By	AUTOKID SUBIC TRADING CORP Lot 28 & 29, Block L, 14 th Streetgate, Perimeter Road, Subic Gateway District, Subic Bay Freeport Zone

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS PABES 301:2020	AMTEC Test
Rated Maximum Power, kW	67.1 ^a	67.1 ^a
Rated Engine Shaft Speed, rpm	2200 ^a	2200 ^a
Maximum PTO Power, kW	57.5	55.73
Percentage PTO Power of the Engine Power, %, minimum	85.0	83.06
Specific Fuel Consumption, g/kW-h, maximum	350	288.03 ^b
At Maximum PTO Power		
Engine Speed, rpm	ND ^c	2127
PTO Shaft Speed, rpm	ND ^c	701
PTO Torque, N-m	ND ^c	760.00
Fuel Consumption, L/h	ND ^c	19.11
Specific Fuel Consumption, g/kW-h	ND ^c	288.03
At 540 rpm PTO Shaft Speed		
Engine Speed, rpm	ND ^c	1653
PTO Torque, N-m	ND ^c	933.50
PTO Power, kW	ND ^c	52.76
Fuel Consumption, L/h	ND ^c	15.72
Specific Fuel Consumption, g/kW-h	ND ^c	250.29

^a The data were obtained from the operator's manual and brochure submitted by the test applicant.

^b Specific fuel consumption at maximum PTO power.

^c Not specified by the test applicant.

Note: ND – No Data

SUMMARY (Cont'n)

Performance Criteria	Requirement as per PNS/BAFS PABES 301:2020	AMTEC Test
Main Gear	1, 2, 3, 4	1, 2, 3, 4, N
Range Gear	Low, Medium, High	Low, Medium, High, Neutral
Shuttle Gear	Forward, Reverse	Forward, Reverse, Neutral
Slowest Forward Speed, kph	1.39 ^a	1.89
Fastest Forward Speed, kph	28.67 ^a	32.09
Radius of Turning Circle, mm (Without Brakes)		
Right Turn	6180 ^a	5530
Left Turn	6130 ^a	5328
Position and/or Draft Implement Controls	Shall be provided	Provided
Three-Point Linkage		
Category based on PTO power at rated rotational frequency of engine	2N/2: 30-92 kW	2N/2
Category based on dimensions of tractor linkage points	2N/2: d ₁ : 25.7+0.2 d ₂ : 28.7+0.3	2N/2
Power-Take-Off Shaft	540 rpm shall have 6 straight splines or 100 rpm shall have 21 involute splines	540 rpm (6 straight splines)
Drawbar Category based on PTO power	2: PTO power ≤115 3: PTO power ≤185 4: PTO power ≤200 5: PTO power ≤500	2, 3, 4, and 5
Drawbar Power, kW	ND ^b	47.93 ^c
Hydraulic Lift Force Capacity, kN	14.86 ^d	22.59 ^e
Wheel Tread Adjustment	Shall be provided	Provided
PTO Casing/Cover	Shall be provided	Provided
Roll-Over-Protective Structure (ROPS)	Shall be provided	Provided
Protective Seat Belt	Shall be provided	Provided

^a Data were obtained from the operator's manual submitted by the test applicant.

^b Not specified by the test applicant.

^c 86% of maximum PTO power.

^d Computed as per Table 8 of PNS/BAFS/PABES 301:2020, requirement if drawbar power is equal to 65 kW or below.

^e Without booster cylinder. 610 mm from the lower link endpoint.

Note: ND – No Data

OBSERVATIONS

1. Operator's manual	
a. Language	English
b. Operator's manual have its own part number and date of issue	Provided
c. Section that provides information that will enable the operator to locate and identify the whereabouts of serial numbers and/or codes of the major components of the machine	Provided
d. Introduction is included in the OM stressing the importance of the information given in the manual.	Provided
e. Attention is drawn to the use of safety alert symbol to highlight information about potential dangers to the user.	Provided
f. Content list is provided to identify the main categories of information in the manual and where they can be found. Location: Right-hand page	Provided
g. Safety signs that appear on the machinery are reproduced in legible size in the operator's manual.	Provided
i. Location of each safety sign on the machine	Provided
ii. Instructions on the need to keep safety signs clear and visible on the equipment	Provided
iii. Instructions to replace safety signs if they are missing or illegible	Provided
iv. Instructions that new equipment components installed during repair shall include the current safety signs specified by the manufacturer and shall be affixed to the replacement component	Not provided
v. Instructions on how to obtain replacement safety signs	Provided

OBSERVATIONS (Cont'n)

h. Operating information with logical instructions for effective operation of the machinery.	Provided
i. Authorized accessories and attachments and its affect in safety, operation and maintenance	Provided
j. Maintenance instructions and schedules are provided	Provided
k. Instructions and information regarding the proper storage of the machinery including precautions to be taken and any tools or special equipment required to prepare the machinery for storage.	Not provided
l. Technical information and instructions for handling, reception, transportation, assembly, installation and initial set-up of the machinery	Provided for transportation only.
m. Relevant dimensions and technical data necessary to assist the operator achieve a higher standard of operational performance and reliability (Specifications of the machinery)	Provided
n. Action to be taken on the completion of the useful life of the machine or its parts, with instructions on dismantling and disposal.	Provided
o. Parts list is included within the operator's manual (if there is no separate parts list or catalogue exists)	Not provided
p. Manuals are available in the language of each country where the machine is sold (Filipino or English)	English
q. Consistent forms of language, spelling, numbering, symbols, etc. are used throughout the manual	Provided
r. All numbers are written in Arabic numerals	Yes
s. Title and number are provided for each table.	Provided
t. Illustrated parts catalogue and repair manual	Not provided
u. Specific parts covered by warranty	Provided

OBSERVATIONS (Cont'n)

2. Previous test reports of similar brand and model	2024 – 0072
3. Manufacturer	INTERNATIONAL TRACTORS LTD. 146022 Jalandhar Road, Hoshiarpur, Punjab, India
4. Serial number	AYYDP1420702MS
5. Failures or abnormalities that may be observed on the machine or its components during and after the operation.	No failures and abnormalities were observed during and after the operation.
6. All tractors shall be equipped with Roll-Over Protective Structures (ROPS) and seat belts.	Provided
7. Seat shall be provided which will adequately support the operator in all working and operating conditions. Adequate and comfortable support and protection for the feet shall also be provided. Seats and ROPS shall conform to the requirements specified in PAES 139 and 140: 2004.	The tractor seat is adjustable and provided with a seat belt. The ROPS is free from manufacturing defects that may be injurious to the tractor operator, free from rust, and painted properly. The ROPS have two posts with a length and width of 1910 mm and 95.91 mm, respectively. The canopy is bolted to the posts.
8. When the PTO is in use, a cover or casing that protects the sides of the PTO shaft shall be fitted. An additional non-rotating casing shall also be provided when the PTO is not in use. This casing shall enclose the PTO shaft completely and be fixed to the tractor body.	A PTO cover was provided. A non-rotating casing that enclosed the PTO shaft in the bottom, left, and right is also provided and fixed to the tractor body.
9. The engine of the tractor shall be equipped with cooling system suitable for tropical operations	The engine is water-cooled.
10. Mechanism that minimize/reduce vibration shall be provided	It was not verified if the machine's vibration was minimized/reduced due to limited technical capabilities to quantify and compare vibration levels during the test operation.
11. The four-wheel tractor shall be free from manufacturing defects that may be detrimental to its operation	No manufacturing defects that may be detrimental to the operation of the four-wheel tractor were observed during the test operation.

OBSERVATIONS (Cont'n)

12. Any uncoated metallic surface shall be free from rust and shall be painted properly	All metallic surfaces are painted properly and free from rust.
13. Appropriate labels, safety symbols, and warnings shall be provided appropriately.	Provided
14. All hot surfaces that may come in contact with the operator shall be provided with heat shield or protective cover.	Provided
15. Attachment of ballast or other weights shall be provided, as necessary.	Provided
16. The four-wheel tractor shall be free from sharp edges and surfaces that may injure the operator. Warning notices shall be provided in accordance with PAES 101:2000.	No sharp edges or surfaces that may injure the operator were observed. Warning notices provided for the tractor were durable and affixed to the machine where parts present danger to the operator. Particular danger or safety warnings were also indicated. Additionally, warning notices are in black text and white, orange, and yellow backgrounds.
17. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Dates of Test	November 12, 2024
Location of Test	Roberto C. Bautista Hall, UPLB, College, Los Baños, Laguna
Items Tested	Operator's Manual, PTO Load Test, Hydraulic Lift, Traveling Speed and Radius of Turning Circle/Area

METHODS OF TEST

The Four-Wheel Tractor was tested based on the PNS/BAFS PABES 302: 2020 – Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE MACHINE

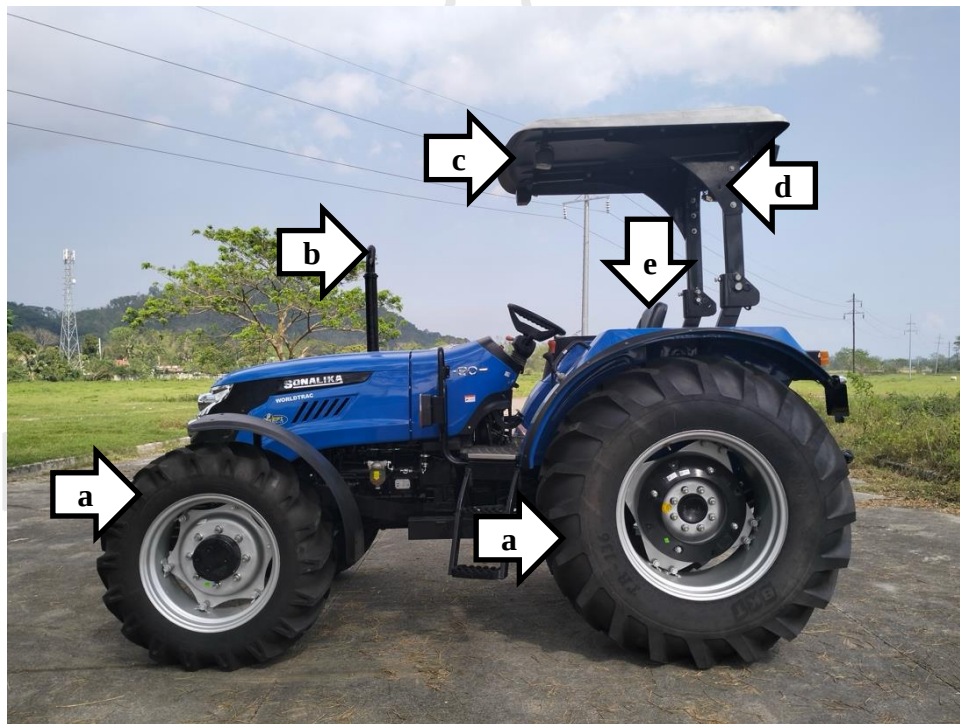


Figure 1. The different parts of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor include the following: (a) Transport wheels, (b) Exhaust, (c) Canopy, (d) Roll-over Protective Structure (ROPS), and (e) Operator's seat.

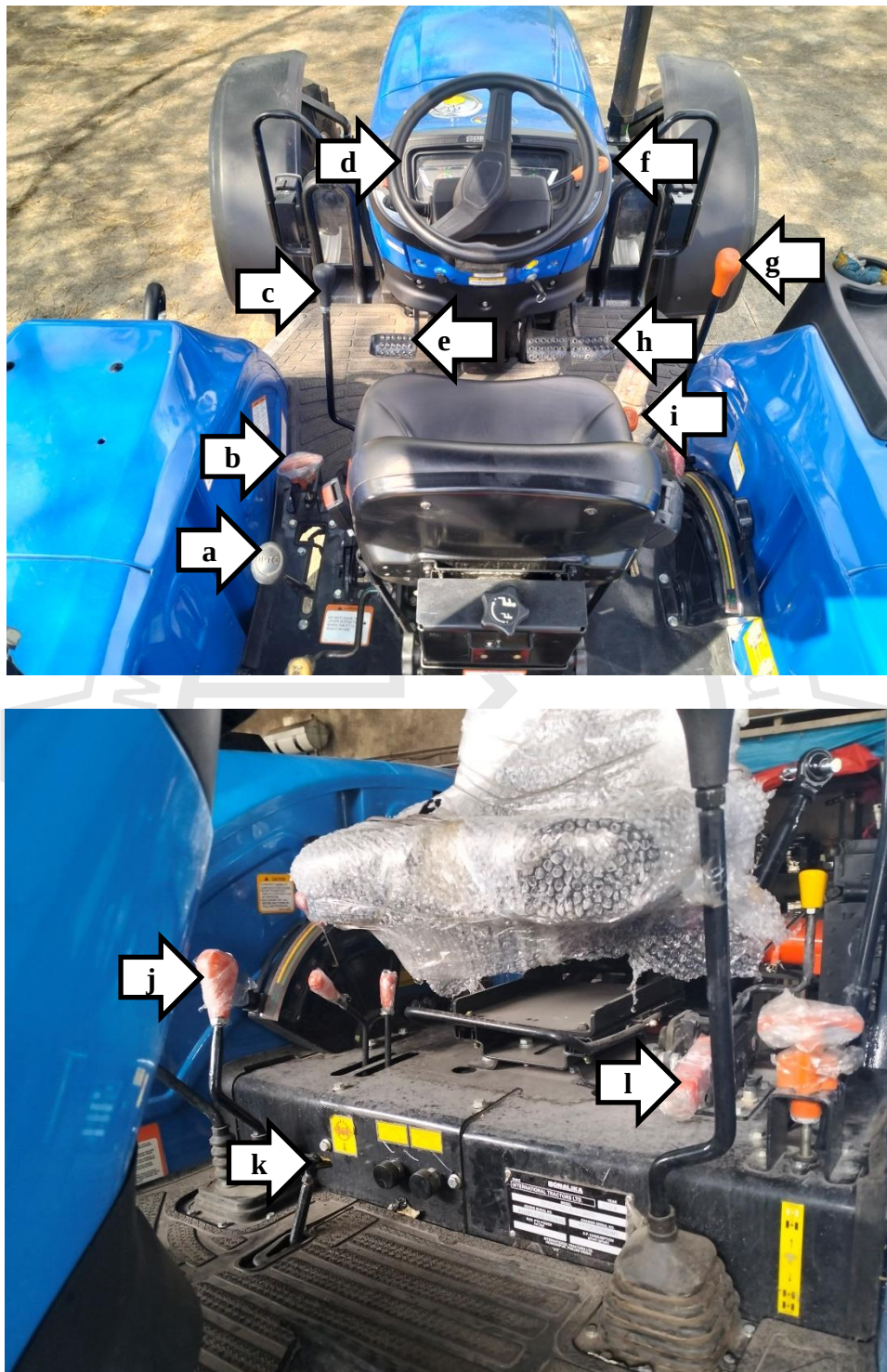
DESCRIPTION OF THE MACHINE (Cont'n)

Figure 2. Controls of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor include the following: (a) PTO clutch lever, (b) Four-wheel drive lever, (c) Shuttle gear shift lever, (d) Steering wheel, (e) Clutch pedal, (f) Hand throttle lever, (g) Main gear shift lever, (h) Brake pedal, (i) Range gearshift lever, (j) Position and draft control lever, (k) Differential lock pedal, (l) Parking brake lever, and (l) Hand brake lever.

SPECIFICATIONS

Item	Manufacturer's Specification ^a	AMTEC Verification
1 Overall dimensions and weight of tractor		
1.1 Length, mm	4106	4160
1.2 Width, mm	1980	2002
1.3 Height, mm	2545	2855
1.4 Dry weight of tractor, kg	3252	3252 ^a
2 Engine		
2.1 Brand	ND	ND ^b
2.2 Make	INTERNATIONAL TRACTORS LTD. India	INTERNATIONAL TRACTORS LTD. India
2.3 Model	4105FLT	4105FLT
2.4 Serial number	4105FLT44C1413474	4105FLT44C1413474
2.5 Output power, kW	67.1	67.1 ^a
2.6 Rated shaft speed, rpm	2200	2200 ^a
2.7 Number of cylinders	4	4
2.8 Bore × stroke, mm	105 × 118	105 × 118 ^a
2.9 Displacement, cm ³	4087	4087 ^a
2.10 Type	Four-stroke, water-cooled, direct injection, turbo charged, four stroke diesel engine	Four-stroke cycle, 4-cylinder, water-cooled, turbo-charged
2.11 Fuel used	Diesel	Diesel
2.12 Governor	ND	ND ^b
2.13 Air cleaner	Dry air cleaner with clogging sensor	Double, dry element
2.14 Lubrication system	ND	ND ^b
2.15 Cooling system	Water-cooled	Water-cooled
2.16 Starting system	ND	Electric starter
2.17 Electrical system	Battery-alternator	Battery-alternator
2.18 Exhaust system	Underhood downdraft silencer, Underhood upward	Vertical muffler
3 Steering system	Power steering	Full hydrostatic

^a The data were obtained from the operator's manual and brochure provided by the test applicant.

^b Not indicated in the operator's manual and brochure submitted by the test applicant

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
4	Power take-off		
4.1	Location	ND	Rear of the tractor
4.2	Type	ND	Type I
4.2.1	Diameter of PTO shaft ends, mm	ND	34.89
4.2.2	Number of splines	6	6
4.2.3	Rated shaft speed, rpm	540/540E	540/540E ^a
4.3	Height above ground, mm	ND	758
4.4	Direction of rotation (viewed from the rear of the tractor)	ND	Clockwise
4.5	Mode of operation		
4.5.1	Speed change	540/540E	Lever-operated (540)
4.5.2	Clutch	Independent and ground PTO (540/540E)	Independent
4.6	Rated PTO Power, kW	57.5	57.73 ^b
5	Ground clearance, mm	ND	425
6	Three-point linkage		
6.1	Category		
6.1.1	Based on PTO power at rated rotational frequency of engine	ND	2N/2
6.1.2	Based on dimensions of tractor linkage point	ND	2N/2
6.2	Length of lift arms, mm	ND	295
6.3	Length of lower links, mm	ND	928
6.4	Horizontal distance between the two lower links, mm	ND	420
6.5	Horizontal distance between the two lift arms endpoints, mm	ND	530
6.6	Length of upper link, mm		
6.6.1	Minimum	ND	580
6.6.2	Maximum	ND	872
6.7	Distance of lower link pivot point to lift rod pivot points on lower links, mm	ND	462

^a The data were obtained from the operator's manual and brochure provided by the test applicant.

^b Based on actual PTO load test.

Note: ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
6.8	Length of lift rods, mm		
6.8.1	Minimum	ND	456
6.8.2	Maximum	ND	620
6.9	Diameter of top pin hole, mm	ND	25.66
6.10	Diameter of lower pin hole, mm	ND	28.90
7	Drawbar ^b		
7.1	Type	ND	Swinging
7.2	Drawbar thickness, mm		
7.2.1	Upper clevis	ND	20.05
7.2.2	Lower clevis	ND	40.09
7.3	Category based on PTO power	ND	2, 3, 4, and 5
7.4	Drawbar width, mm	ND	Upper: 60.03 Lower: 70.25
7.5	Pin hole diameter, mm	ND	Upper hole: 33.26 Lower hole: 33.53
7.6	Pin diameter, mm	ND	33.23
7.7	F, mm	ND	Upper hole: 37.84 Lower hole: 39.36
7.8	G, mm	ND	257
7.9	Height, mm	ND	75.54
7.10	Throat depth, mm	ND	70.34
7.11	End radius of drawbar and clevis, mm	ND	55.07
7.12	W, mm	ND	NA

^a The test applicant did not submit complete machine specifications.

^b Based on Table 6 of PNS/BAFS PABES 301:2020.

Note: NA – Not applicable (The tractor's drawbar has no chamfer), ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
8	Transmission system		
8.1	Main clutch and PTO clutch	Double clutch, mechanical	Dry type single-plate clutch
8.2	Transmission gears		
8.2.1	Type	Synchromesh (12+12)	Synchromesh
8.2.2	Main gear	1, 2, 3, 4	1, 2, 3, 4, N
8.2.3	Range gear	Low, Medium, High	Low, Medium, High, Neutral
8.2.4	Shuttle gear	Forward, Reverse	Forward, Reverse, Neutral
8.3	Differential lock	Standard, foot operated	Mechanical pedal
9	Tire		
9.1	Type	ND	Bias-ply
9.2	Front tire size, D × W, mm	315 × 610	315 × 610
9.3	Rear tire size, D × W, mm	467 × 762	467 × 762
9.4	Wheelbase, mm	2345	2330
9.5	Wheel tread		
9.5.1	Front tread, mm	1525 to 1940	1690
9.5.2	Rear tread, mm	1455 to 1845	1508
10	Brake system		
10.1	Type according to manner of applying braking force	ND	ND ^b
10.2	Type according to manner of transmitting the force from the controls	ND	Mechanical
10.3	Parking brake	ND	Parking brake lever

^a The data were obtained from the operator's manual provided by the test applicant.

^b The engine was not disassembled during verification.

Note: ND – No Data

RESULTS

Table 1. Radius of turning circle/area of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

Item	Without Brake		With Brake	
	Right Turn	Left Turn	Right Turn	Left Turn
Radius of Turning Circle, mm	5530	5328	5113	4683
Radius of Turning Area, mm	5683	5480	5265	4835

Table 2. Traveling speed of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor at different gear settings.

Range Gear Settings	Main Gear Settings	Shuttle Gear Settings	
		Forward (kph)	Reverse (kph)
Low	First	1.89	1.55
	Second	2.79	2.31
	Third	3.84	3.16
	Fourth	5.44	4.71
Medium	First	4.43	3.76
	Second	6.42	5.62
	Third	9.16	7.83
	Fourth	13.81	11.26
High	First	12.48	10.23
	Second	18.42	15.08
	Third	25.37	21.33
	Fourth	32.09	31.03

RESULTS (Cont'n)

Table 3. Test conditions during the power take-off load test of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

Temperature, °C	35.9
Relative Humidity, %	61.6

Table 4. Power Take-Off Load Test data of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

Speed		PTO		Fuel Consumption, L/h	Specific Fuel Consumption, g/kW-h
Engine, rpm	PTO, rpm	Torque, N-m	Power, kW		
2378	783	70	5.70	8.16	1203.51
2308	760	420	33.42	13.82	347.36
2242	738	707	54.56	19.43	299.16
2182	718	733	55.13	19.34	294.59
2127	701	760	55.73	19.11	288.03
2067	680	782	55.63	18.81	284.05
2005	659	804	55.48	18.54	280.63
1948	640	829	55.52	17.95	271.60
1892	620	849	55.12	17.59	267.99
1836	603	868	54.72	17.22	264.36
1771	581	897	54.60	16.84	259.02
1712	562	915	53.81	16.47	257.12
1653	540	934	52.76	15.72	250.29
1598	521	950	51.83	15.44	250.22

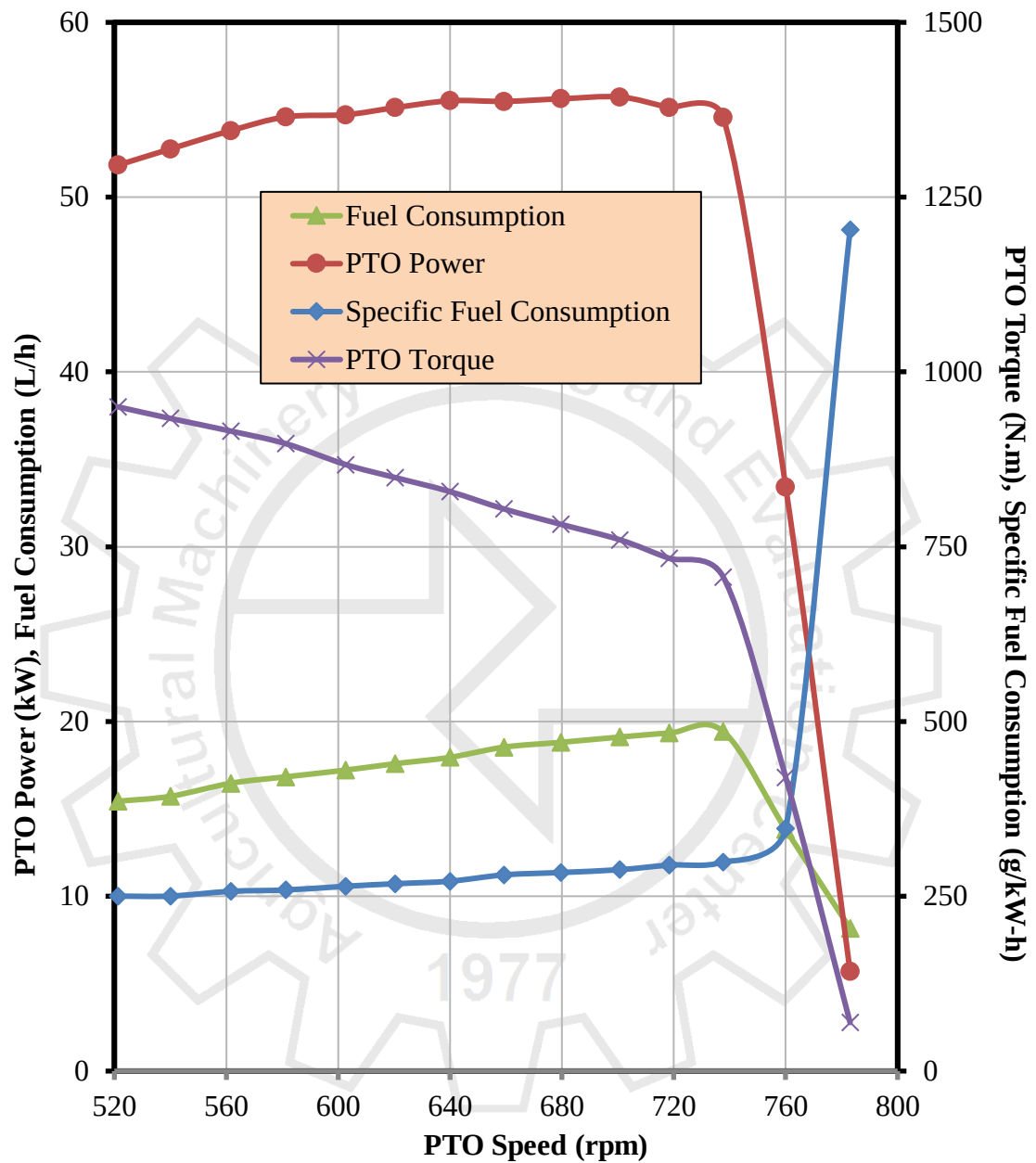
RESULTS (Cont'n)

Figure 3. Performance Characteristics Curves of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

RESULTS (Cont'n)

Table 5. Hydraulic Lift Test data of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor without booster cylinder.

Trial	On the Frame	
	kg	kN
1	2309.2	22.64
2	2301.3	22.57
3	2302.4	22.58
4	2306.8	22.62
5	2301.0	22.56
Average	2304.1	22.59

Note: Without booster cylinder. 610 mm from the lower link endpoint.

Table 6. Drawbar power and lift force capacity of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

Drawbar power, kW	47.93 ^a
Hydraulic lift force capacity, kN	14.86 ^b

^a 86% of maximum PTO power.

^b Computed as per Table 8 of PNS/BAFS/PABES 301:2020, requirement if drawbar power is equal to 65 kW or below.



Figure 4. Hydraulic lift setup of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

OBSERVATIONS



Figure 5. Nameplate of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.



Figure 6. Markings and labeling on the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

OBSERVATIONS (Cont'n)

Figure 7. Safety warnings of the SONALIKA WORLDTRAC 90RX (MST) Four-Wheel Tractor.

Tested by:


CHRISTOPHER S. JAPONES
Test Engineer

Verified by:


FATIMA JOY R. AYING
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

11 DEC 2024
Date

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